

Approved by:
June 3, 2024

R. Vaugois



Type of Product

ZGS-Z-500

Serial No.:JM2402036

Drawing No.:ZGS500-231124-02

ANHUI JIEZOU NEW ENERGY CO.,LTD.

1 Description			
Name of Product	Three-phase Pad Mount Transformer		
Type of Product	ZGS-Z-500		
Serial No.	JM2402036		
Drawing No.	ZGS500-231124-02		
2 Technical Specification			
Power Rating	500kVA	Cooling Type	KNAN
Rated Voltage	7.2/0.6kV	Number of Phases	3 Phases
Rated Current	40.09/481.14A	Rated Frequency	60 Hz
Vector Group	Dyn1		
Basic Insulation Level	HV	LI/AC 95/26 kV	
	LV	LI/AC 30/10 kV	
3 Applicable standards			
ANSI/IEEE C57.12.00 General Requirements for Liquid-Immersed Distribution, Power and Regulating Transformers ANSI/IEEE C57.12.90 Test Code for Liquid-Immersed Distribution, Power and Regulating Transformers ANSI/IEEE C57.12.80 Standard Terminology for Power and Distribution Transformers ANSI/IEEE C57.100 IEEE Standard Test Procedure for Thermal Evaluation of Liquid-Immersed Distribution and Power Transformers ANSI/IEEE C57.136 IEEE Guide For Sound Level Abatement & Determination For Liquid-Immersed Power Transformers & Shunt Reactors Rated Over 500 KVA ASTM D 6871 Standard Specification for Natural (Vegetable Oil) Ester Fluids Used in Electrical Apparatus AWS American Welding Society NEMA National Electrical Manufacturers Association NFPA TO National Fire Protection Association National Electrical Code SSPC The Society for Protective Coatings Paint Application Specification NO. 2 CSA C227.4:21 National Standard of Canada			

4 Items and results of tests

4.1 Measurement of voltage ratio and check of vector group

H.V.		L.V.		Calculated Ratio	Deviation of Measured Voltage Ratio (%)			Tolerance	Vector Group
Tap Position	Voltage (V)	Tap Position	Voltage (V)		H1H2/X1X2	H2H3/X2X3	H3H1/X3X1		
1	7560	/	600	12.600	0.04	0.04	0.04	≤±0.5%	Dyn1
2	7380			12.300	0.06	0.06	0.06		
3	7200			12.000	0.07	0.07	0.07		
4	7020			11.700	0.09	0.09	0.09		
5	6840			11.400	0.10	0.10	0.10		

4.2 Measurement of winding resistance

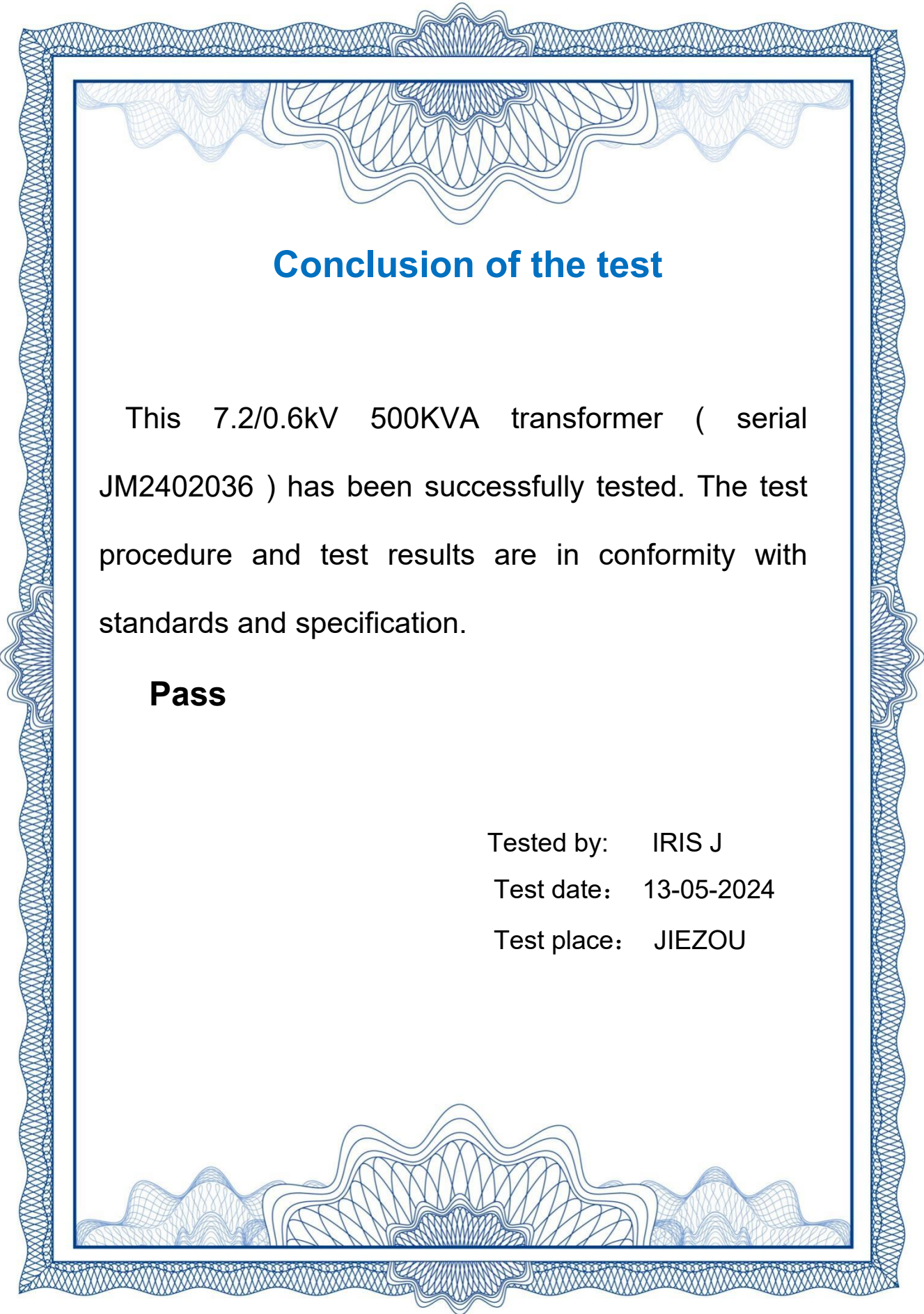
Average Oil Temperature:19°C

Tap Position	H.V. Windings (Ω)			Unbalance Rate
	H1H2	H2H3	H3H1	Measured
1	1.227	1.228	1.228	0.1%
2	1.196	1.196	1.196	0%
3	1.166	1.165	1.165	0.1%
4	1.134	1.133	1.133	0.1%
5	1.103	1.100	1.103	0.3%
Tap Position	L.V. Windings (Ω)			Unbalance Rate
	X1X2	X2X3	X3X1	Measured
/	0.004839	0.004830	0.004870	0.8%
	X1X0	X2X0	X3X0	Measured
/	0.002505	0.002484	0.002514	1.2%

4.3 Measurement of Insulation Resistance								
Relative Humidity:54%,Top Oil Temperature:19℃								
Measured Between	1min Insulation Resistance (MΩ)		Voltage (V)					
H.V.~L.V. &Earth	> 2500		2500					
L.V.~H.V.& Earth	> 2500							
H.V. & L.V.~Earth	> 2500							
4.4 Applied Voltage Test (AV)								
Measured Between	Test Voltage (kV)	Test Frequency (Hz)	Duration (s)	Result				
H.V.	26	50	60	pass				
L.V.	10	50	60	pass				
4.5 Induced Voltage Withstand Test(IVW)								
Tap Position	Tested Winding	Supplied Terminal	Supplied Voltage (kV)	Induced Voltage (kV)	Induced Multiple	Test Frequency (Hz)	Duration (S)	Result
3	H.V.	L.V.	1.2	14.4	2	150	48	Pass

4.6 Measurement of No-load Current and No-load Loss								
Average Oil Temperature: 19℃								
Test Voltage (kV)		Test Current			No load loss shall be corrected to degree C			
Mean	R.M.S	I0(A)	I0(%)		Corrected Loss (P0)			
			Required	Measured				
0.6	0.6007	1.24	/	0.26	502W			
4.7 Measurement of Short-circuit Impedance and Load Loss								
【H.V.~L.V.】								
Tap Position		Test Current I(A)	Test Voltage U(kV)	Impedance Base:400kVA		Load losses shall be corrected to 85 degree C		
H.V.	L.V.			Required	Tolerance	Uk(%)	Zk(Ω)	Load Loss
3	/	40.01	0.2609	/	/	3.70	/	5825W

4.8 Tank Pressure (Leakage) Test				
Test Method	Applied Pressure (kPa)	Duration of Test (h)	Release Pressure (kpa)	Result
Tank Pressure	15	12	15	No leaks



Conclusion of the test

This 7.2/0.6kV 500KVA transformer (serial JM2402036) has been successfully tested. The test procedure and test results are in conformity with standards and specification.

Pass

Tested by: IRIS J

Test date: 13-05-2024

Test place: JIEZOU